

A molecular ferroelectric thin film of imidazolium perchlorate on silicon

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Molecular ferroelectrics have garnered significant attention due to their structural tunability, low synthesis temperature, and high flexibility. Herein, we successfully synthesized imidazole perchlorate (ImClO₄) single crystals and high-quality, highly-oriented thin films on Si substrates. These films demonstrated a high inverse piezoelectric coefficient of 55.7 pm/V. Two types of domain bands were observed: type-I bands tilted ~60° relative to the horizontal axis, and type-II bands positioned perpendicular to the horizontal axis. Under a +20 V bias, type-I bands showed a reduction and detachment of 180° domain walls to form a needle-like domain. It extended toward the band boundary after applying -20 V bias, which grew along the boundary upon contact. In contrast, type-II bands showed straight domain wall motion and displayed a higher piezoresponse than type-I bands. The growth of high quality molecular ferroelectric thin films on Si substrates paves the way for the development of on-chip devices.

Molecular ferroelectrics are a type of ferroelectric materials made up of small organic molecules with weak intermolecular forces like hydrogen bonding and van der Waals forces^{1–5}. They have been considered promising candidates for applications in flexible electronics, with advantages such as low manufacturing temperature, low cost, use of non-toxic elements, high flexibility, and lightweight when compared to the traditional polar oxides. Previous research has shown that molecular ferroelectric materials can achieve high electric properties comparable to or even higher than oxide ferroelectrics with molecule design^{6–13}. Among these materials, Imidazole perchlorate (ImClO₄) has been widely investigated, with high Curie temperature (up to 373 K¹⁴, close to BaTiO₃), facile sample preparation, relatively high spontaneous polarization, and low coercive field, etc^{14–19}. In recent years, various ImClO₄ thin films have been successfully synthesized, showing excellent electrical properties, including superior electromechanical coupling¹⁶, ionically controlled ferroelectric behavior¹⁷, strong electrocaloric effect¹⁸, tunable electroresistance and electro-optic effects¹⁹.

These studies have highlighted the potential applications of ImClO₄-based ferroelectric materials and devices. However, there are

two main obstacles to the practical applications of molecular ferroelectric materials in electronic devices. First, integrating molecular thin films into Si-based devices (such as transistors, memories, and sensors) is challenging. Unlike the inorganic oxides that can be deposited layer-by-layer on a Si substrate through pulsed laser deposition, molecular beam epitaxy, or chemical vapor deposition^{20,21}, the growth of high-quality highly oriented molecular ferroelectric thin films on Si is challenging due to the significant differences in chemical and physical properties between ImClO₄ and Si. Second, the deterministic control of the ferroelectric domain switching, as well as the fundamental understanding of the domain switching kinetics, are still lacking for molecular ferroelectrics, which is critical to their use in electronic devices^{22–24}.

Results and discussion

Herein, we have successfully synthesized high-quality, large-scale, and highly oriented ImClO₄ single crystals using a facile solution-evaporation method. Details of the single crystal growth procedure are introduced in the Methods and depicted in Fig. S1. The structure

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characterizations and electrical properties measurements of the as-grown ImClO_4 single crystal are presented in Fig. 1. Transparent single crystals with a size of up to $1.2 \times 0.8 \times 0.4 \text{ cm}^3$ have been successfully synthesized (Fig. 1a). The X-ray diffraction (XRD) analysis of the powder sample, obtained from the single crystal, shows peak intensities and positions that are consistent with data reported in a previous study¹⁸. Meanwhile, only two XRD diffraction peaks at 16.40° and 33.12° can be identified for the single crystal, corresponding to the (100) and (200) lattice planes, respectively¹⁸. This suggests that the samples are highly oriented along the [100] direction with good crystallinity, in contrast to the single crystal grown by Li et al.¹⁸ which was mainly oriented along the [110] direction and exhibited multiple diffraction peaks. Differential Scanning Calorimetry (DSC) characterization is further performed to unveil the ferroelectric transition temperatures, as shown in Fig. 1b. Endothermic/exothermic peaks can be seen at 373.4 K/371.6 K, showing that the ferroelectric phase transition is a first-order type transition with a Curie temperature of $\sim 372 \text{ K}$, consistent with the previous report¹⁴. The FTIR transmission peaks at 624 cm^{-1} , 763 cm^{-1} , and 1090 cm^{-1} correspond to the out-of-plane vibration, in-plane deformation, and symmetric stretching of the $[\text{ClO}_4]^-$ groups, respectively, while the peak at 1586 cm^{-1} represents the symmetric stretching of imidazolium cations from the $\text{C}=\text{C}$ bond¹⁸.

The morphology and crystal plane of an ImClO_4 single crystal were analyzed using high-resolution transmission electron microscopy (HRTEM), as illustrated in Fig. 1d–f. The black and white contrast with a size of $\sim 100 \text{ nm}$ can be seen, indicating distinct ferroelectric domains separated by domain walls. Figure 1e provides a magnified view of the single crystal, revealing periodic lattice planes such as the (110) and (200) crystal planes with lattice spacings of 0.45 nm and 0.27 nm , respectively. This is further supported by the selected area electron diffraction (SAED), where the (200) , (110) , and (110) diffraction points can be identified (Fig. 1f). Furthermore, the electric properties of the single crystal were examined. Figure 1g displays the dielectric constants and losses across different frequencies at ambient temperature. With increasing frequency, the dielectric constant and dielectric loss both show a monotonic decreasing trend. From 1 kHz to 1 MHz , the relative dielectric permittivity decreases from 119.86 to 31.3 , while the dielectric loss also decreases from 9.13 to 0.14 . A rectangular P–E loop with a saturation polarization of $1.81 \mu\text{C}/\text{cm}^2$ at 294 K can be obtained with a very low coercive field of $\sim 8.5 \text{ kV}/\text{cm}$, as shown in Fig. 1h. Moreover, the presence of two current peaks near the coercive field further confirms the ferroelectricity of the ImClO_4 single crystals. The I–V curve of the single crystal is also measured (Fig. S2), demonstrating a low leakage current comparable to the previous reports^{17,19}. Additionally, the

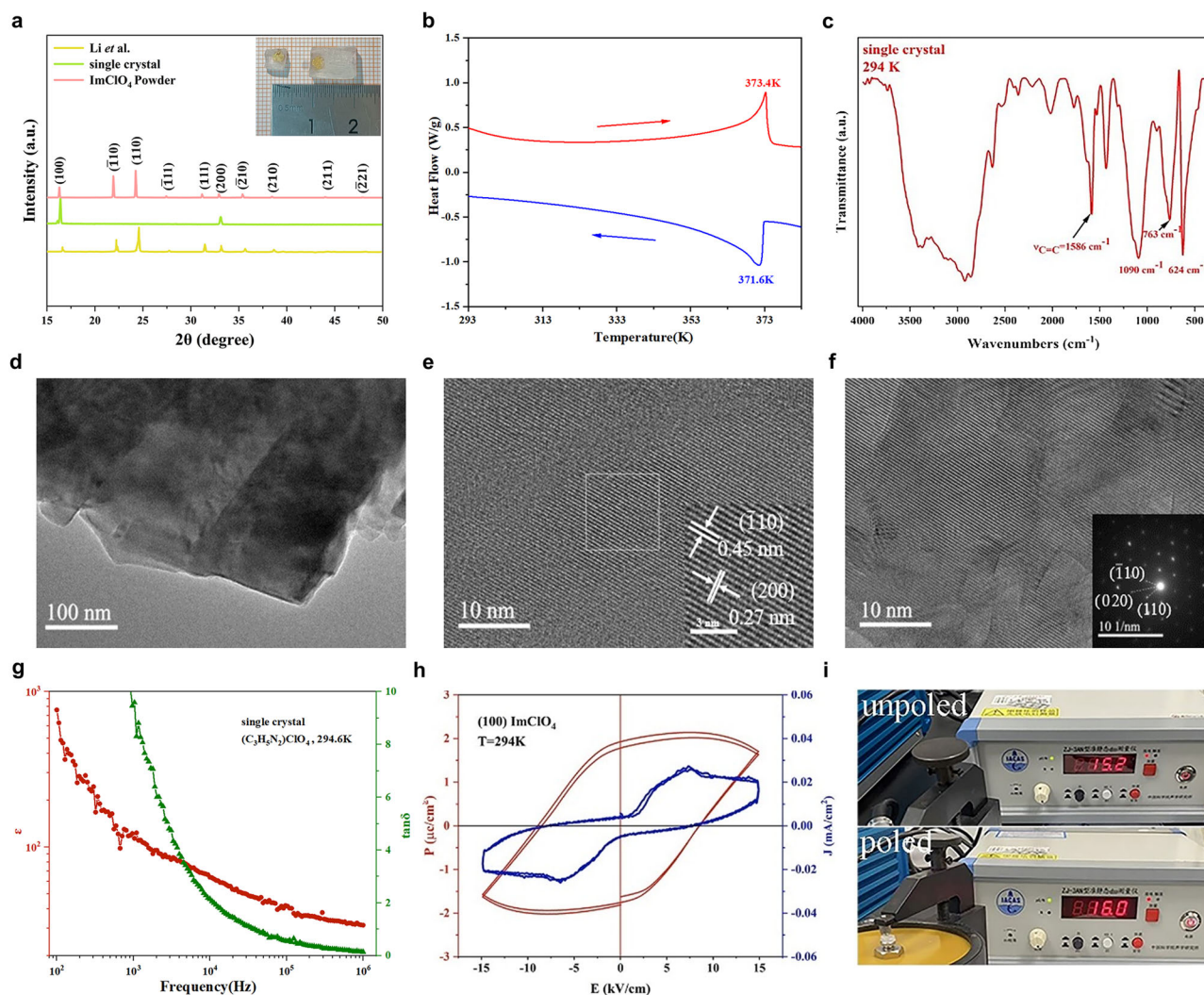


Fig. 1 | Structure characterizations and electric properties of the ImClO_4 bulk single crystal. **a** XRD patterns of the ImClO_4 single crystal and powder. Inset: crystal morphology. **b** Differential scanning calorimetry (DSC) measurements. **c** Fourier Transform Infrared (FTIR) spectra. **d–f** The HRTEM images. Inset: fast

Fourier transform of the HRTEM image. **g** The dielectric constant (ϵ') and the dielectric loss ($\tan \delta$) at various frequencies. **h** The typical polarization-electric field (P-E) hysteresis loop and the current density curve. **i** Piezoelectric properties tested before and after polarization.

unpoled ImClO₄ single crystals show a high direct piezoelectric coefficient (d_{33}) of 15.2 pC/N at room temperature, which slightly increases to 16 pC/N after poling (Fig. 1i). These measurements demonstrate that the highly (100)-oriented single crystals exhibit excellent electric properties.

The ImClO₄ thin films with a thickness down to 1 μm were prepared using the spin coating method. The ImClO₄ single crystals were dissolved in water to obtain super saturated solutions, which are then coated on Si single crystal wafers after heating and annealing (details in Methods). In this process, sodium alginate (SA) is added to the solvent to aid the nucleation process and promote oriented growth. SA, being a water-soluble biomolecule, is commonly used in the assembly of nanostructures^{25,26}. The structural characterizations of the as-grown thin films were performed. The surface morphologies of the thin films are characterized using an optical microscope (Fig. S3). Fig. S3a–c displays herringbone-like structures in ImClO₄ films prepared using SA addition. Interestingly, as the rotation speed increases, the size of the bone increases (~50 μm @3000 r/min, ~100 μm @6000 r/min, and >200 μm @8000 r/min). At high rotation speeds, such as 8000 r/min, the side branches can be distinguished from the branches on the stem. Notably, at a rotation speed of 6000 r/min, the angles between a stem and the branches are 60° on the left and 87° on the right (Fig. S3b), in line with previous reports¹⁹. This clearly demonstrates that the domain size and the ratio between the stem and branches can be modulated by adjusting the rotation speed. In contrast, without the SA additive, diverse morphologies emerge, including droplet-like, herringbone-like, and leaf-like structures (Fig. S3d–f). The thin films prepared without SA exhibit significantly reduced surface coverage compared to those with SA, underscoring SA's crucial role in the nucleation and growth of ImClO₄ crystals. Further enhancement of film coverage can be achieved by employing additional spin coating processes, as evidenced in Fig. S3g, h.

The growth mechanisms of the thin film with/without SA are shown in Fig. S4. In the case with SA, the SA particles initially form on the Si surface due to their strong adhesion with Si. During annealing, the ImClO₄ crystals nucleate on the SA particles and grow along preferred orientations, resulting in herringbone-like structures. Heterogeneous nucleation significantly lowers the nucleation barrier, resulting in good crystallinity. Conversely, without SA, the nucleation barrier is higher since ImClO₄ is generally incompatible with Si, leading to the formation of droplet-like and leaf-like structures.

The XRD patterns in Fig. 2a show the ImClO₄ films prepared by spin coating at different rotation speeds (3000–8000 r/min). The diffraction peaks at 22° dominate, confirming the highly (110)-oriented growth of the ImClO₄ films assisted with SA. To evaluate the quality of the thin films, synchrotron-based Grazing Incidence Wide Angle X-ray Scattering (GIWAXS^{27–29}) was used (Fig. 2b). Two major diffraction peaks can be identified, along with several weak rings, indicating high crystalline quality and slight surface misalignment. The (110) diffraction peak shows the highest intensity, with a slightly weaker (100) peak, consistent with the XRD measurements.

The domain structure is further characterized by piezoelectric force microscopy (PFM). Figure 2c shows the topography of the ImClO₄ film prepared by spin coating at 6000 r/min, a region with a flat surface is selected, with a dimension of ~4 μm . The lateral amplitude and phase are presented in Fig. 2d, e. A clear domain wall with a 180° phase contrast can be identified, with decreasing amplitude in the vicinity of the domain wall. A butterfly displacement loop can be seen, corresponding to the ferroelectric switching with a hysteric 180° phase shift (Fig. 2f). Besides, the inverse piezoelectric coefficient of the ImClO₄ film can be calculated as -55.7 pm/V, which is among the highest reported inverse piezoelectric response values in this system¹⁹. The ferroelectric property is further investigated with second harmonic generation (Fig. 2g), a 100 × 100 μm^2 spatial image was taken by point-by-point scanning across the sample region, showing well-

organized herringbone-like domain boundary, with a size of several micrometers. The whole film exhibited rather high SHG intensity around the bone-like regions, indicating the formation of ferroelectric phase with high nonlinear optical response^{30,31}. It is also interesting to note that there's a huge variation of the SHG signal in the different regions, showing the possibility to directly modulate the second-order susceptibility through electric field³¹. Previously, the high SHG intensity has also been observed in TMCM-CdCl₂ nanoplates³². We then measure the ferroelectric polarization by measuring the SHG dependency of light polarization, as shown in Fig. 2h–i). In horizontal, vertical as well as couple setups, two-fold symmetry was observed pointing to around 10°, corresponding to the polarization direction with low symmetry.

The Fourier-transformed infrared spectroscopy (FTIR) image under attenuated total reflection-infrared (ATR-IR) mode for ImClO₄ films at different rotation speeds is shown in Fig. S5. The FTIR reflection peaks at 736 cm^{-1} , 1026 cm^{-1} , 1573 cm^{-1} , and 3250 cm^{-1} can be identified, corresponding to the C-H bending, C-N stretching, C=C stretching, and primary amine functional groups³³, with subtle differences observed for films with different rotation speeds. Meanwhile, the FTIR peak at 2300 cm^{-1} corresponds to the O=C=O stretching mode for CO₂³⁴, which exhibits variations with different rotation speeds, due to the presence of air in the testing environment. Moreover, it is worth noting that the peak positions for the thin films show a slight decrease in wave numbers compared to the bulk single crystal (e.g., 736 cm^{-1} , 1026 cm^{-1} , and 1573 cm^{-1} for film *vs.* 763 cm^{-1} , 1090 cm^{-1} , and 1586 cm^{-1} for bulk). This phenomenon can be attributed to the higher surface area for the thin films that lowers the bond strength, thereby lowering the FTIR frequency.

Figure 3 illustrates the polar domains in SA-assisted ImClO₄ thin films, as analyzed by the piezoelectric force microscopy (PFM). To improve the film coverage, the thin films were subjected to a double coating process at a rotation speed of 6000 r/min, with the corresponding optical image presented in Fig. S3g. Two distinct structures were identified from the topology mapping, corresponding to two different morphologies labeled as type-I and type-II domain bands. The topology, out-of-plane PFM (VPFM) amplitude, VPFM phase, in-plane PFM (LPFM) amplitude, and LPFM phase for the type-I band are displayed in Fig. 3a–e, respectively. In this scenario, the domain boundaries tilt ~60° with respect to the horizontal axis (*x*-axis in the image), while the domain walls are perpendicular to the boundaries. The majority of domain walls exhibit 180° changes in both in-plane and out-of-plane phases, indicating that they are 180° domain walls. For type-I band, some other domain walls show only 180° shifts for out-of-plane phase, indicating that they can be 71° domain walls. The type-II band is oriented parallel or perpendicular to the horizontal axis (Fig. 3f–j). Both 180° domain walls and 109° domain walls can be identified from the in-plane and out-of-plane amplitude and phase changes. The correlation between the in-plane and out-of-plane amplitude is weak, indicating the formation of pure in-plane 180° domain walls, consistent with the domain walls calculated from the phase-field simulations. Additionally, the domain morphologies for the thin films with different rotational speeds are compared in Fig. S6. It is evident that while the domain shapes are similar for all the three cases, the domains are generally smaller with increasing rotational speed. This can be attributed to the decrease in film thickness with increasing rotational speed, leading to a decrease in domain sizes according to the Kittel's law.

The in-situ switching study for type-I band is performed by applying a local bias through a static PFM tip located in the center of the image, as illustrated in Fig. 4. The schematic diagram of the PFM system can be found in Fig. S7. The tomography of a selected area (4 × 4 μm^2) of typical type-I band is given in Fig. 4a, where the domain boundary tilts 60° relative to the horizontal axis. The initial domain structure can be seen from the vertical and lateral PFM images in Fig. 4b–e. The 180° domain walls are perpendicular to the boundary,

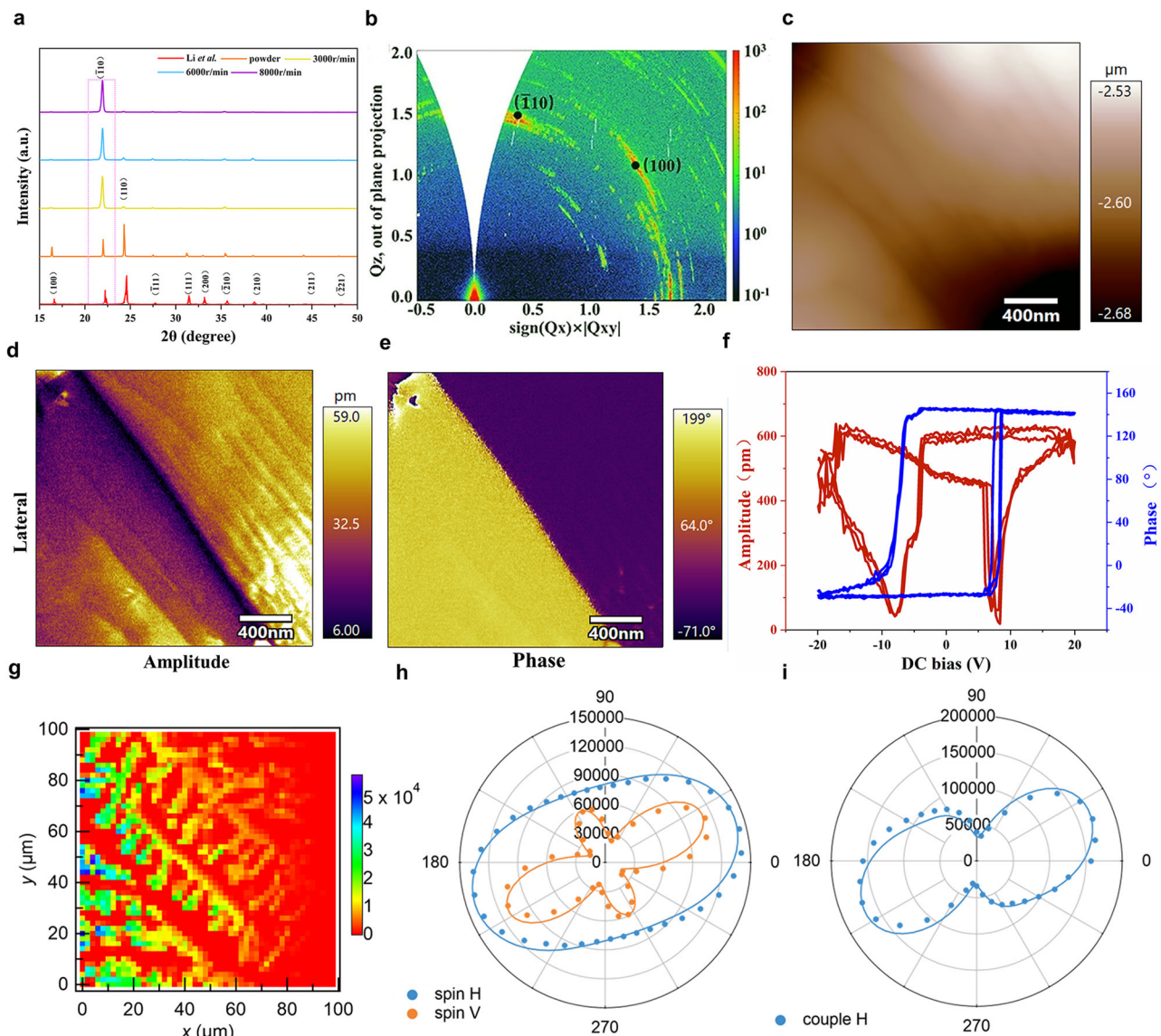


Fig. 2 | Structure characterizations, ferroelectric domain characterizations, and second harmonic generation (SHG) measurements of the highly oriented ImClO₄ thin film prepared with 1% SA. a XRD pattern of ImClO₄ thin film prepared at different rotational speeds. **b** The grazing-incidence wide-angle X-ray scattering (GIWAXS) pattern. **c–e** PFM Topography, lateral amplitude, and phase images. **f** The

local PFM hysteresis loops: the amplitude (red) and phase (blue) signals as functions of bias voltage. **g** The spatial SHG intensity mapping in a 100 × 100 μm² sample region shows herringbone-like intensity contrast. **h, i** The polarization-dependent SHG spectra for horizontal, vertical, and couple configurations (dots). The lines are the corresponding fitting results.

forming a superdomain structure with a band size of 1 μm. In the same region, the in-plane and out-of-plane response magnitude is similar, suggesting the presence of both in-plane and out-of-plane polarization. The domain walls are highly curved towards the band edge, indicating that they can be charged domain walls. The formation of charged domain walls in weak ferroelectrics has been investigated previously^{35,36}. Following the application of a positive bias of 20 V (effective electric field ~200 kV/cm) for 5 s in the center of the region, the domain band with strong piezoresponse shrinks gradually, disconnecting from the bottom domain to form submicron needle-like domains (Fig. 4f–i). Simultaneously, the small initial 180° domains were completely switched. Subsequently, an opposite bias of -20 V is applied, the needle domain extends towards the band boundary after 5 s, while the needle size is almost constant (Fig. 4j–m). This suggests that the needle edge is in a highly charged state and is more responsive to the applied electric field. The entire switching process can be repeated, where the application of a positive bias leads to the

formation of needle-like domains again, while a reverse bias could invert the switching. Notably, when the pulse duration is extended to 10 seconds under -20 V, the needle extends along the band boundary, potentially increasing in size up to 1 μm.

To further investigate the switching kinetics for type-I band, we employed the phase-field simulations to reveal the domain switching after applying an electric field, as shown in Fig. 4n–w. The initial domain configuration is illustrated in Fig. 4n, where the (110)-oriented ImClO₄ thin film has a rhombohedral phase at room temperature, consistent with previous XRD results. A biaxial coherent strain of 0 is set to simulate a fully relaxed ImClO₄ thin film on Si. Details of the phase-field simulations can be found in the Methods. The planar view of the stable domain structure is presented in Fig. 4o, with arrows indicating the polarization direction. The domain wall configurations follow the mechanical compatibility criteria³⁷, and the phase-field simulation results closely match those observed in the experiment. Subsequently, we applied a bias of 20 V along the out-of-plane

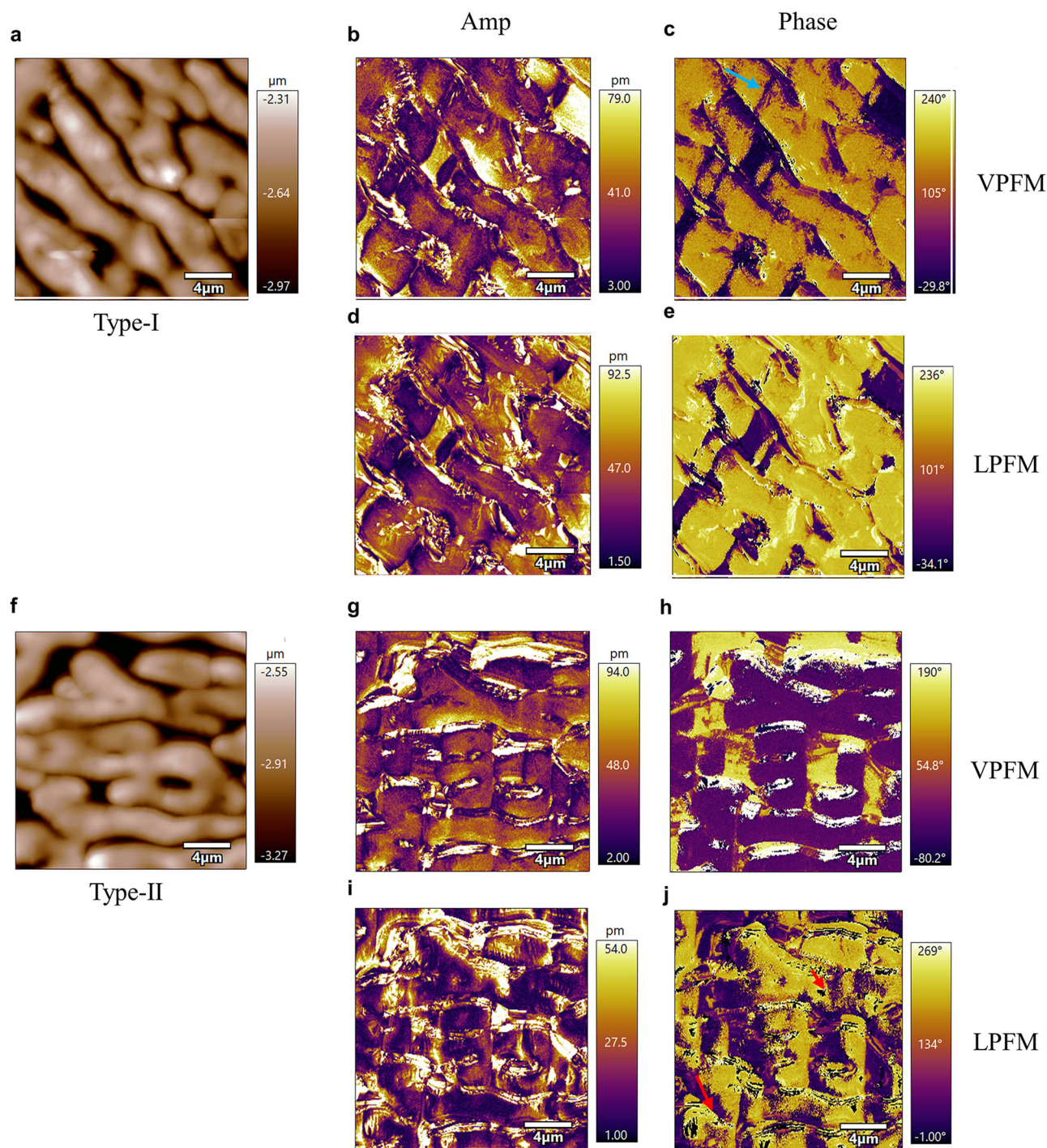


Fig. 3 | The PFM images for ImClO_4 thin films prepared by spin coating at 6000 r/min. a, f Topography. **b, g** VPFM amplitude. **c, h** VPFM phase. **d, i** LPFM amplitude. **e, j** LPFM phase. The blue arrow in (c) indicates a 71° domain wall, while the red arrows in (j) show 109° domain walls.

direction of films to switch these domains. The green domain band shrinks continuously, while the other domains are relatively immobile, as shown in Fig. 4p–r, disconnecting from the bottom red domain and forming needle-like domains. Figure 4s provides a three-dimensional view of the equilibrium domain configuration after 20000 TS. An opposite bias (–20 V) was applied next, causing the green domain to extend back toward the band boundary, which then expanded laterally as they met each other, as depicted in Fig. 4u–w. The whole switching kinetics is consistent with the previous PFM studies. This study indicates that for type-I band, only the domains with both high in-plane and out-of-plane polarization are responsive

to the external field, which will reduce the overall dielectric and piezoelectric responses.

Next, we proceed to understand the switching kinetics of the type-II band (Fig. 5). The topography of the selected area is given in Fig. 5a. The initial PFM images indicate the coexistence of 180° and 109° domain walls that are perpendicular/horizontal or orient 45° to the horizontal axis (Fig. 5b–e). The 180° domain wall is very straight, suggesting that it is a non-charged domain wall. After applying +20 V for 5 s, the 180° domain wall moves to the left side, while the 109° domain wall is less mobile, only extending with the motion of the 180° domain wall (Fig. 5f–i). Subsequently,

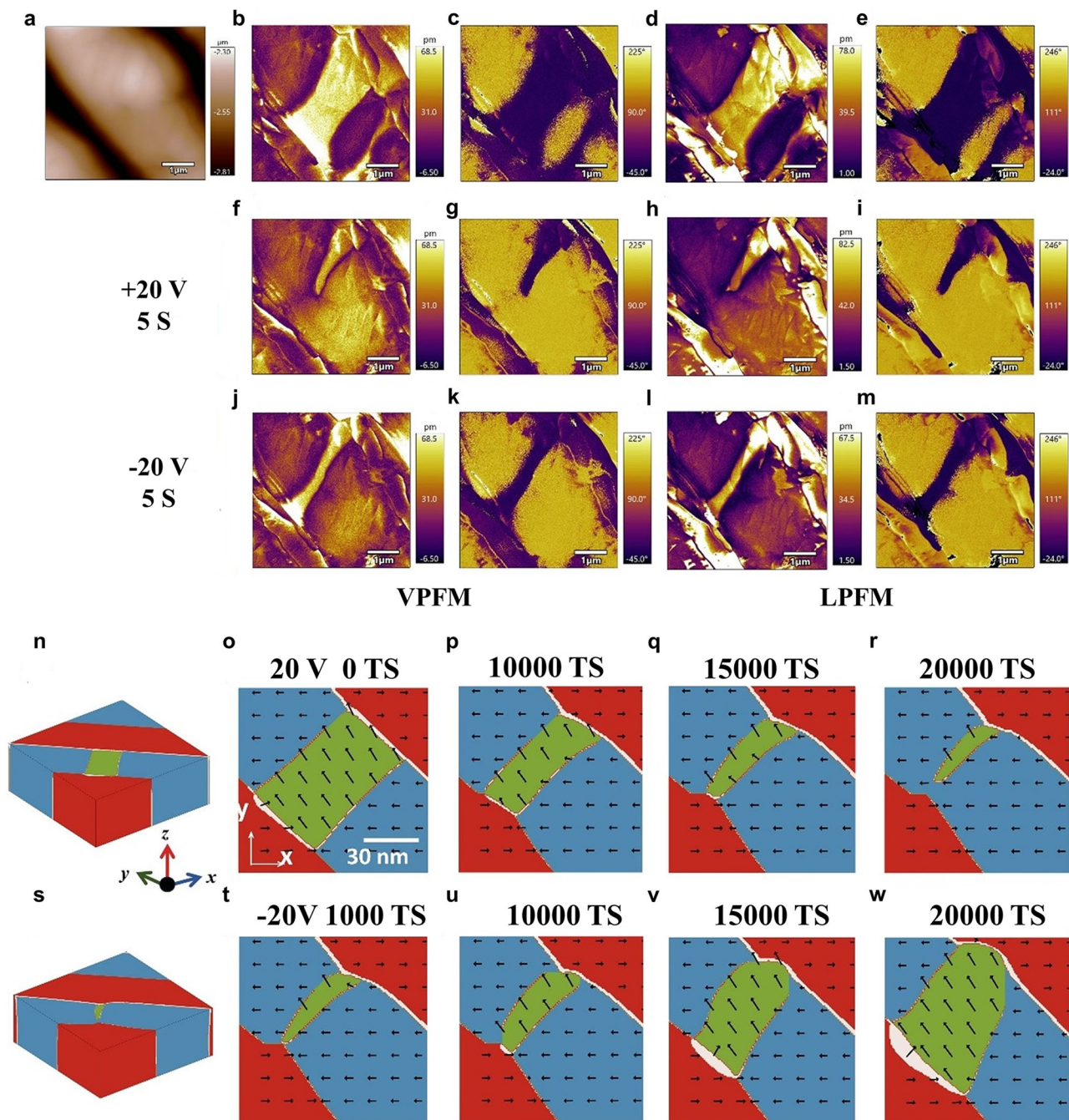


Fig. 4 | The in situ PFM switching and corresponding phase-field simulations for type-I band. **a** Topography images with an area of $2.5 \times 2.5 \mu\text{m}^2$. **b–e** Initial VPFM amplitude, phase, LPFM amplitude, and phase images, respectively. **f–i** VPFM amplitude, phase, LPFM amplitude, and phase images after applying a bias of +20 V for 5 s in the center of the area, respectively. **j–m** VPFM amplitude, phase, LPFM amplitude, and phase after applying a bias of -20 V for 5 s in the center of the area,

respectively. **n** Three-dimensional view of the stabilized domain configuration from phase-field simulations. **o–r** Temporal evolution of domains on the top surface under 20 V. The black arrows indicate the polarization directions. **s** Domain configurations under 20 V after 20000 TS (Timesteps). **t–w** Temporal evolution of domains on the top surface under applying a bias of -20 V.

when a reverse bias is applied for 5 s, the 180° domain wall moves to the right side for 200 nm, much shorter than the initial displacement of 500 nm after applying +20 V (Fig. 5j–m). This suggests that the field driving motion for the 180° domain wall could cause fatigue for this system.

The kinetic evolution of domains under a 20 V bias is further elucidated through phase-field simulations in Fig. 5n–w. The initial domain configuration is illustrated in Fig. 5n, and the polarization vectors of the orange and blue domains appear to be oriented tail-to-tail or head-to-

head looking down the film, as depicted in Fig. 5o. However, from the XZ plane in Fig. S8, the out-of-plane polarization vectors of these two domains are arranged side-by-side, forming alternating positive and negative polarization regions separated by straight and non-charged 180° domain walls. On the two sides of the 180° domain wall, both inplane and out-of-plane polarization exists. The orange domain is situated on the left side of the domain wall (exhibiting an upward out-of-plane polarization component), while the blue (with a downward out-of-plane polarization component) is on the right side.

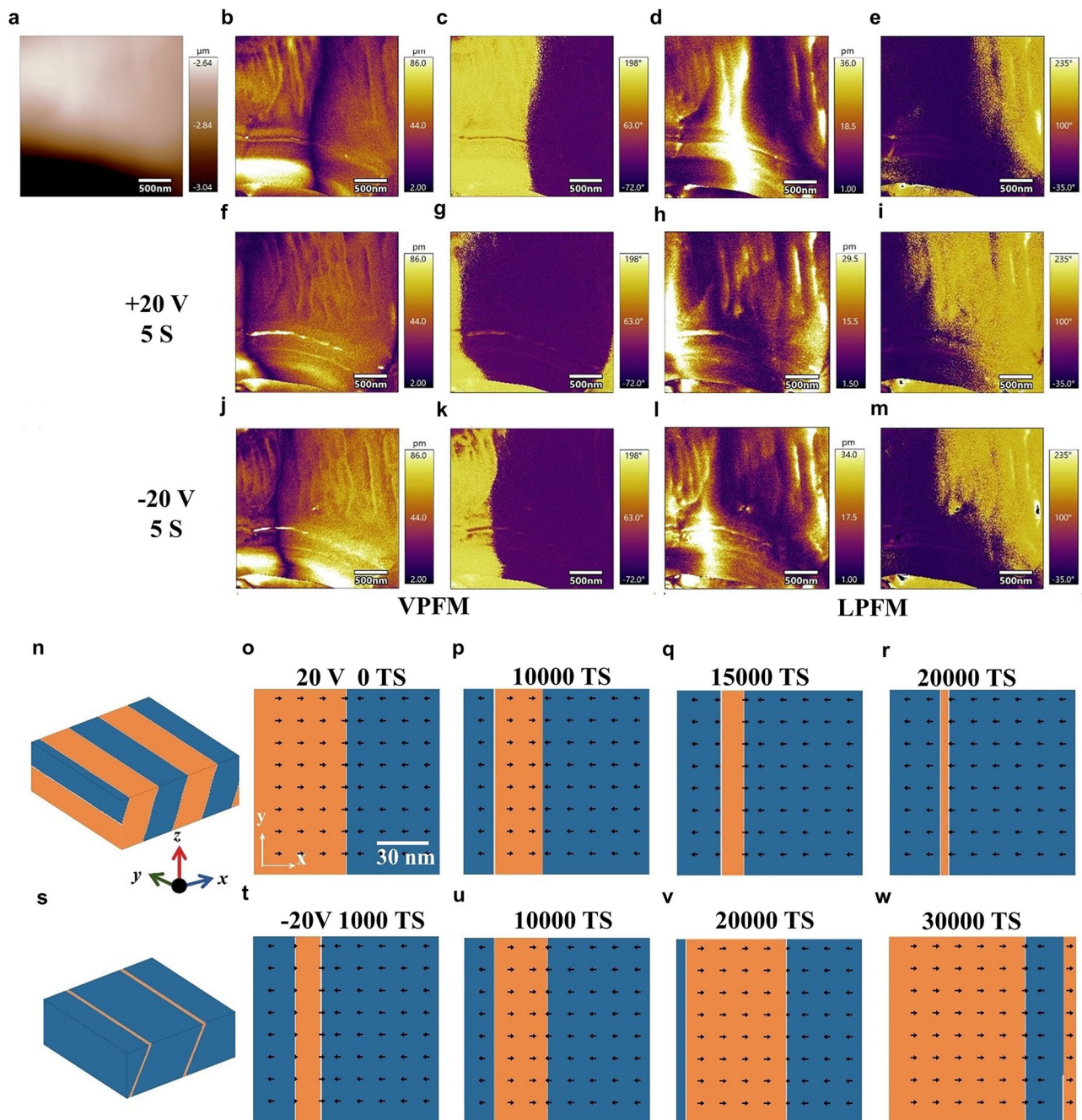


Fig. 5 | The in situ PFM switching and corresponding phase-field simulations for type-II band. **a** Topography images with $2.5 \times 2.5 \mu\text{m}^2$. **b–e** Initial VPFM amplitude, phase, LPFM amplitude, and phase images, respectively. **f–i** VPFM amplitude, phase, LPFM amplitude, and phase images after applying a bias of +20 V for 5 s in the center of the area, respectively. **j–m** VPFM amplitude, phase, LPFM amplitude, and phase after applying a bias of -20 V for 5 s in the center of the area, respectively.

n Three-dimensional view of the stabilized domain configuration from phase-field simulations. **o–r** Temporal evolution of inplane polarization under 20 V after different timesteps. **s** Three-dimensional view of the final domain configuration after applying 20 V. **t–w** Temporal evolution of inplane polarization under -20 V after 1000, 10000, 20000, and 30000 TS, respectively.

With a vertical downward electric field applied under a 20 V bias (Fig. 5o–r), the orange domain begins to contract as both domain walls progress horizontally, whereas the blue domain expands. After 10,000 TS, the orange domains are reduced to half their initial size. At 20,000 TS, the orange domain appears as merely a thin line. Applying a reverse voltage of the same magnitude (-20 V) in Fig. 5s–w, the orange domain starts to expand, and the domain walls on both sides advance toward the blue domain, which is consistent with the experimental data.

Fig. S9 compares the switching ratio of the two bands at different time steps from phase-field simulations. Notably, for type-II band, the switching ratio is higher than for type-I band. The butterfly loop for the two bands is compared in Fig. S10, showing a higher piezoresponse for type-II band than for type-I band. This can be understood that the switching ratio for type-I band is smaller than for type-II band, which will give rise to a lower mechanical response under the same applied voltage.

Conclusions

In summary, we have developed a facile solution-evaporation method to grow large-scale (~1 cm in size) high-quality (100)-oriented ImClO₄ single crystals. They exhibit good ferroelectricity (spontaneous polarization -1.81 μC/cm² at 294 K), low coercive field (~8.5 kV/cm), high Curie temperature (~372 K), and high piezoelectric coefficient (15.2 pC/N for unpoled crystal). Then, highly (110)-oriented ImClO₄ thin films are grown on a Si substrate with the sol-gel spin-coating and annealing method, using sodium alginate as the nucleation additive. XRD, optical microscopy, and PFM characterizations are performed to unveil the crystal structure, morphology, piezoelectric, and ferroelectric properties of the thin films. A high inverse piezoelectric coefficient (55.7 pm/V) is obtained in this film.

Two types of domain bands (in the size of a few microns) are observed: the type-I band tilts ~60° with respect to the horizontal axis, while the type-II band is perpendicular to the horizontal axis. Both domain bands are formed mainly by 180° domain walls, with minor 109° domain walls. Most of the domain walls in the type-I band are curved, charged domain walls, while the 180° domain walls in the type-II band are straight, noncharged domain walls. The switching kinetics for the two bands are investigated with PFM. After applying +20 V for 5 s, the 180° domain walls in the type-I band shrink first, then disconnect from the band boundary, forming a needle-like domain with a size of ~100 nm. At the same time, the smaller domains (~100 nm) are entirely switched. The needle-like domain will extend toward the band boundary after an inverse bias is applied (~20 V), starting from the needle tip point, and expand along the band boundary after touching the boundary. Whereas for the type-II domain band, the 180° domain walls are more mobile than the 109° domain walls, which displace ~500 nm after applying +20 V. The displacement of the domain wall is much shorter when a negative bias is applied for the same amount of time, starting from the positively poled sample.

This research presents a simple approach to grow highly-oriented, large-scale molecular ferroelectric single crystals and thin films. The SA-assisted thin film growth methodology can be extended to the production of other high-quality molecular ferroelectric thin films. The work offers a comprehensive and detailed examination of domain wall kinetics in molecular ferroelectrics, revealing that molecular ferroelectric thin film properties can be modulated by fine-tuning growth conditions and employing domain engineering. The thin films are grown directly on a Si substrate, offering potential advantages for integrating molecular-ferroelectric-based electronic devices on a chip. We aim to stimulate broader interest in enhancing the quality of molecular ferroelectric thin films and in advancing the practical design of molecular-ferroelectric-based electronic devices.

Methods

Crystal growth and film preparation

(100)-oriented molecular ferroelectric single crystals with sizes up to 1.2 × 0.8 × 0.4 cm were prepared by slow evaporation of equimolar imidazole (500 mmol) and perchloric acid (500 mmol) solutions at room temperature. To get ImClO₄ films, ImClO₄ crystals obtained by evaporating the solution and sodium alginate were dissolved in deionized water and stirred for 24 hours to form a solution, wherein the proportion of sodium alginate mixed was 1 wt%. After using a plasma cleaner to carry out hydrophilic treatment on the surface of the single-crystal silicon substrate, the previously prepared solution was spin-coated on the substrate surface evenly at a speed of 3000–8000 r/min through a spin-coater. A series of molecular ferroelectric ImClO₄ films of different thicknesses can be prepared by drying the films in an oven at a low temperature of 80 °C for 1 h and annealing the films again at 120 °C for 1 h. The specific synthesis process of crystal growth and film preparation can be found in Fig. S1 in the Supporting Information.

DSC, SHG, XRD and GIWAXS measurements

Differential scanning calorimetry (DSC) measurements were carried out by using a DSC Q100 instrument under the nitrogen atmosphere, where the single crystal of ImClO₄ (2.94 mg) was heated and cooled with a rate of 10 K/min in the temperature ranges of 288–394 K. The X-ray diffraction (XRD) data of ImClO₄ single crystal were obtained from Rigaku fabricated Miniflex-600c in the 2θ range of 15°–50° with a step size of 0.02°. The grazing incident XRD (GIXRD) data of ImClO₄ film at different rotational speeds were collected to detect the intensity of crystal planes on the Bruker D8 Discover using Cu Kα radiation (λ = 1.54059 Å) with a scan rate of 10°/min. Powder X-ray diffraction (PXRD) was obtained from Rigaku fabricated Miniflex-600c and was analyzed by depositing ImClO₄ powder in a glass substrate with 0.5 mm depth, and the detecting angles started from 2θ = 15° to 2θ = 50°, with 0.02° increment. GIWAXS measurements were performed at Xeuss 3.0 with high-brightness micro-focus spot solid-state Cu target light source (λ = 1.542 Å). The SHG measurements were performed using a home-made SHG microscopy system. A Ti:sapphire mode-locked femtosecond laser (MaiTai SP, Spectra-Physics) was used to generate the 800 nm excitation laser beam (80 MHz, 35 fs). The laser was directed onto the sample at normal incidence, focused by a 50X objective lens (NA = 0.55) to an approximate diameter of 1 μm. The generated second harmonic light was subsequently collected by a photomultiplier tube. For spatial SHG mapping, the focal point was scanned by a pair of galvanometers. For the spin measurements, a half-wave plate was used to rotate the fundamental light polarization direction, while keeping the analyzer (a Glan-Taylor prism) fixed at either horizontal or vertical direction. For the couple measurements, the half-wave plate and the analyzer were rotated in parallel. More details about the equipment and the spectral imaging method can be found in reference¹³.

Dielectric and ferroelectric measurements

The instrument BALAB DMS2000 was used to detect the frequency-dependent dielectric constants of ImClO₄ single crystals under the frequency range from 100 Hz to 1 MHz with an applied electric field of 1 V. The longitudinal piezoelectric coefficient (d₃₃) was tested by the quasi-static meter (ZJ-3A, Chinese Academy of Sciences, China) under the low-frequency constant force (0.25 N) at room temperature. The P-E loops and current density of ImClO₄ single crystals were carried out using a ferroelectric test system (Premier II, Radiant Technologies Inc., USA). The ferroelectric polarization imaging and local switching measurements on the bulk crystal surface were carried out using a PFM (MFP-3D, Asylum Research).

FTIR and Optical microscopic measurements

The Fourier-transformed infrared spectroscopy (FTIR, Vertex 70) under transmission mode and attenuated total reflection-infrared (ATR-IR) mode were utilized to characterize the bulk crystals and films, respectively. The optical microscopic images were obtained by a probe stage (HCP421V-MPS).

Phase-field simulations

In the current phase-field model, the order parameter is selected to representing the spontaneous polarization vector $\mathbf{P} = (P_1, P_2, P_3)$. The time-dependent evolution of the domain structure is acquired by solving the time-dependent Ginzburg-Landau equations:

$$\frac{dP_i}{dt} = -\frac{L\delta F_{\text{tot}}}{\delta P_i}, (i=1,2,3) \quad (1)$$

where L is the kinetic coefficient related to the domain wall mobility; The total free energy F_{tot} can be expressed by the volume integral of the bulk Landau free energy, elastic energy, electrostatic energy and

polarization gradient energy:

$$F_{\text{tot}} = \iiint (f_{\text{Land}} + f_{\text{elas}} + f_{\text{elec}} + f_{\text{grad}}) dV \quad (2)$$

The Landau energy density is calculated by a sixth order Landau-Devonshire polynomial:

$$\begin{aligned} f_{\text{Land}} = & \alpha_1(p_1^2 + p_2^2 + p_3^2) + \alpha_{11}(p_1^4 + p_2^4 + p_3^4) \\ & + \alpha_{12}(p_1^2 + p_2^2 + p_3^2)(p_1^2 + p_2^2 + p_3^2) + \alpha_{111}(p_1^6 + p_2^6 + p_3^6) \\ & + \alpha_{112}[p_1^4(p_2^2 + p_3^2) + p_2^4(p_1^2 + p_3^2) + p_3^4(p_1^2 + p_2^2)] \\ & + \alpha_{123}p_1^2p_2^2p_3^2 \end{aligned} \quad (3)$$

where α_1 , α_{11} , α_{12} , α_{111} , α_{112} and α_{123} are the Landau coefficients.

The elastic energy density f_{elas} is given by:

$$f_{\text{elas}} = \frac{1}{2} C_{ijkl} (\varepsilon_{ij} - \varepsilon_{ij}^0) (\varepsilon_{kl} - \varepsilon_{kl}^0) \quad (4)$$

in which C_{ijkl} is the elastic stiffness tensor, ε_{ij} is the elastic strain and ε_{ij}^0 is the eigenstrain due to the ferroelectric phase transition. The eigenstrain ε_{ij}^0 is given by

$$\varepsilon_{ij}^0 = Q_{ijkl} P_k P_l \quad (5)$$

where Q_{ijkl} is the electrostriction coefficient tensor.

The electrostatic energy density f_{elec} is described by

$$f_{\text{elec}} = -\frac{1}{2} \varepsilon_0 \kappa_{ij} E_i E_j - E_i P_i \quad (6)$$

where ε_0 is the dielectric permittivity of vacuum and κ_{ij} is the background dielectric constant tensor of the ferroelectric film, and $\kappa_{11} = \kappa_{22} = \kappa_{33} = 40$ is set in this study. The local electric field E_i can be obtained by $E_i = -\nabla_i \varphi$. The closed boundary conditions are configured to ensure that the electric potential φ remains at zero, while applying potentials to the film-substrate interface and the top surface of the thin film, respectively.

The gradient energy density can be formulated as

$$\begin{aligned} f_{\text{grad}} = & \frac{1}{2} G_{11}(p_{1,1}^2 + p_{2,2}^2 + p_{3,3}^2) + G_{12}(p_{1,1}p_{2,2}p_{1,1}p_{3,3} + p_{2,2}p_{3,3}) \\ & + \frac{1}{2} G_{44}[(p_{1,2} + p_{2,1})^2 + (p_{2,3} + p_{3,2})^2 + (p_{1,3} + p_{3,1})^2] \\ & + \frac{1}{2} G'_{44}[(p_{1,2} + p_{2,1})^2 + (p_{2,3} + p_{3,2})^2 + (p_{1,3} + p_{3,1})^2] \end{aligned} \quad (7)$$

where G_{11} , G_{12} , G_{44} and G'_{44} are the gradient energy coefficients, and P_{ij} stands for the spatial differential of P_i , $P_{ij} = \frac{\partial P_i}{\partial x_j}$. The gradient energy coefficients are chosen as $\frac{G_{11}}{G_{110}} = 0.6$, $\frac{G_{12}}{G_{110}} = 0$, $\frac{G_{44}}{G_{110}} = \frac{G'_{44}}{G_{110}} = 0.3$, where $G_{110} = 1.73 \times 10^{10} \text{ C}^{-2} \text{ m}^4 \text{ N}$.

All the parameters employed in this work are taken from refs. 18,38, and listed in Table S1. The simulation system is discretized into a three-dimensional mesh with dimensions of $256 \times 256 \times 150$, where each grid corresponds to a single unit cell. A periodic boundary condition is employed for the dimensions within the plane, while an out-of-plane superposition method is utilized. This out-of-plane direction is composed of 30 grids representing the substrate, 100 grids for the thin ImClO_4 films, and an additional 20 layers of air.

The local coordinate system (x,y,z) with axes along [100], [010], and [001] directions, and the global coordinate system (x',y',z') with axes along [001], [110], and $[\bar{1}10]$ directions are used to describe the $(\bar{1}10)$ -oriented thin film. The transformation matrix t_{ij} can move the

polarization vectors from the local coordinate system (x,y,z) to the global coordinate system (x',y',z') .

$$t_{ij} = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 100 & & \end{bmatrix}$$

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used are available within this manuscript and Supplementary Information. Further information can be acquired from the corresponding authors upon reasonable request.

Code availability

The phase-field simulation results in this work were obtained using the software package Mu. [PRO](#).

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Author contributions

These authors contributed equally: C.Z., X.L., and Z.H. conceived the idea and designed the project. C.Z. fabricated the samples. X.L. performed phase-field simulations. W.L. carried out SHG measurements and analysis. C.Z., X.L., W.L., T.C., F.L. Y.H., and Z.H. analyzed the data. Y.H., Q.L., Y.W., and Z.H. supervised the project. All authors contributed to the discussion, analysis, and manuscript preparation.

Competing interests

The authors declare no competing interest.

Additional information

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